

Final Exam Statistical Mechanics

19 January 2024

Welcome to the final exam of Statistical Mechanics of the GEMIF. The exam covers all the topics we have seen during the course.

The final mark for the course will be $= \max\{0.40 \text{ midterm} + 0.60 \text{ final}, 1.00 \text{ final}\}$

Question 1 [1.5 points]

Consider N magnetic indistinguishable particles with magnetic moment μ that can occupy a reticular substrate with $N_1 + N_2$ sites of types 1 and 2, with $N_1 \geq N$ and $N_2 \geq N$. On the type 1 sites acts a localized magnetic field equal to H , so that the energy of a particle occupying one of these sites is $\epsilon = -\mu H < 0$. Particles occupying type 2 sites have zero energy.

- (a) Find the number of microstates $\Omega(E, N)$ corresponding to an energy value equal to $E = n\epsilon$, with n being the number of particles located on magnetic sites.

Consider a modified scenario where the N particles are on a lattice of $N_1 + N_2 = N$ sites of types 1 and 2, respectively. In this scenario, we can turn on and off the magnetic field independently for each type 1 site. Suppose we turn the magnetic field on in n of the type 1 sites, so that particles occupying these sites have energy $\epsilon = -\mu H$ and the remaining particles have zero energy.

- (b) Find the number of microstates $\Omega(E, N)$ corresponding to energy $E = n\epsilon$, with n being the number of particles occupying N_1 sites with applied magnetic field..

Are the answers to both questions the same? Explain why they are/ are not the same.

Solution Question 1:

- (a) Particles are indistinguishable but sites are distinguishable. Therefore, we need to count the number of combinations of N_1 sites into groups of n (i.e. occupied sites with energy ϵ) and the number of combinations of N_2 sites into groups of $N - n$:

$$\Omega(E, N) = \binom{N_1}{n} \cdot \binom{N_2}{N-n} = \binom{N_1}{E/\epsilon} \cdot \binom{N_2}{N-E/\epsilon} \quad (1)$$

- (b) In this case all sites are occupied. Therefore, we only need to take into account the combinations of N_1 sites in groups of n :

$$\Omega(E, N) = \binom{N_1}{n} = \binom{N_1}{E/\epsilon} \quad (2)$$

the difference comes from the fact that all N_2 sites are occupied, so that in the second scenario $\binom{N_2}{N_2} = 1$.

Question 2:

Consider a system comprising two identical, on-interacting particles, with no spin degeneracy, that can be in two possible states with energies $\epsilon_0 = 0$ and $\epsilon_1 = \epsilon > 0$.

- (a) Write the canonical partition function for Bose-Einstein, Fermi-Dirac, and classical particle statistics.
- (b) Compute the probability of finding the two particles in the ground state for each one of the cases above. For which statistics is this probability the largest?

Solution Question 2:

Partition function:

B-E There are three possible states: (ϵ_0, ϵ_0) , (ϵ_0, ϵ_1) , and (ϵ_1, ϵ_1) , therefore:

$$Z_{BE} = 1 + e^{-\beta\epsilon} + e^{-2\beta\epsilon} \quad (3)$$

F-D There is one possible state: (ϵ_0, ϵ_1) , therefore:

$$Z_{FD} = e^{-\beta\epsilon} \quad (4)$$

C In the classical limit, for non-interacting identical particles we have that $Z = \frac{1}{N!} Z_1^N$, with Z_1 being the partition function for a single particle. Therefore

$$Z_{\text{Classical}} = \frac{(1 + e^{-\beta\epsilon})^2}{2!} \quad (5)$$

Question 3:

In the mean field approximation for the Ising model in 1D, each spin is assumed to feel an average field due to its neighbors. Consider the 1D Ising model with Hamiltonian

$$\mathcal{H} = -J \sum_{i=1}^N s_i s_{i+1} - H \sum_{i=1}^N s_i$$

where $s_i = \pm 1$, J is the interaction strength, and H is the magnetic field. In this approximation, the average spin $\langle s \rangle$ is determined self-consistently as:

$$\langle s \rangle = \tanh(\beta(H + J\langle s \rangle))$$

where $\beta = \frac{1}{k_B T}$ with k_B being the Boltzmann constant and T the temperature.

- (a) Find the value of the correlation function $\langle s_i s_j \rangle$, in the mean field approximation.
- (b) Discuss if this value of the correlation function $\langle s_i s_j \rangle$, will be the same if we do not use the mean field approximation.

Solution Question 3:

In mean field theory we assume that fluctuations are negligible, then the correlation between two spins is approximated by the product of their average values, given that they will act as independent variables: $\langle s_i s_j \rangle \approx \langle s \rangle^2 = \tanh^2(\beta(H + J\langle s \rangle))$. If we do not apply mean field, the correlation will be different given that the homogeneity/independence assumption will not hold

Question 4:

The Landau-de Gennes free energy, used to model the isotropic-nematic transition in nematic forming materials, is given by:

$$F(S, T) = \frac{1}{2}A(T - T_c)S^2 - \frac{1}{3}BS^3 + \frac{1}{4}CS^4$$

where S is the order parameter, and A , B , and C are positive constants independent of temperature. Note that C must be positive to ensure that the free energy does not diverge to minus infinity for large values of S , consistent with the truncation of the series expansion. For the time being, B is assumed to be positive. The Landau-de Gennes free energy represents the essential feature of the isotropic-nematic transition and is often used as a model free energy for this transition.

- (a) According to $F(S, T)$, find the critical temperature T_c at which the nematic transition takes place. This is achieved computing the values of the order parameter at which the Landau-de Gennes free energy vanishes, at the critical point.

Solution Question 4:

Let's take a look at $F(S, T)$. This function has extrema as determined by the condition

$$\frac{\partial F}{\partial S} = A(T - T_c)S - BS^2 + CS^3 = 0 \quad (3)$$

and hence has as extrema always $S = 0$ and

$$S_{\min} = \frac{B \pm \sqrt{B^2 - 4AC(T - T_c)}}{2C}$$

So, when

$$B^2 - 4AC(T - T_c) > 0 \quad \text{or} \quad T - T_c < \frac{B^2}{4AC}$$

two new solutions appear, beside $S = 0$. At this T , the free-energy of the new minimum is still higher than the free energy at $S = 0$.

Substituting Eq. 3 ($\frac{dF}{dS} = 0$), re-written as

$$A(T - T_c) = BS_{\min} - CS_{\min}^2$$

in the free energy expression, it becomes possible to calculate the value of the free energy at the minimum (for $S = S_{\min}$) as

$$\begin{aligned} F(S_{\min}) &= \frac{1}{2}(BS_{\min} - CS_{\min}^2)S_{\min}^2 - \frac{1}{3}BS_{\min}^3 + \frac{1}{4}CS_{\min}^4 \\ &= S_{\min}^3 \left[\left(\frac{1}{2} - \frac{1}{3}\right)B + S_{\min} \left(-\frac{1}{2} + \frac{1}{4}\right)C \right] \\ &= S_{\min}^3 \left(\frac{1}{6}B - S_{\min} \frac{1}{4}C \right) \end{aligned}$$

the free energy in the minimum is zero when $S_{\min} = \frac{2B}{3C}$ and $A(T - T_c) = \frac{2B^2}{9C}$.

Problem 1:

Consider a set of N independent particles moving in 1D on a segment $0 < x < L$. The single-particle Hamiltonian is given by:

$$H = \frac{p^2}{2m} - \alpha \ln \left(\frac{x}{x_0} \right) \quad (6)$$

where p is the momentum, m is the mass of the particles, and α and x_0 are positive constants.

- Compute the single-particle partition function Z_1 .
- Determine the free Hemholtz energy F of the gas.

- (c) Determine the force Ψ that the particles exert against the ends of the segment, given by

$$\Psi = -\left.\frac{\partial F}{\partial L}\right|_{T,N} \quad (7)$$

- (d) Find the behavior of the force Ψ for very low temperatures.

Solution Problem 1:

- (a) We can express the single-particle partition function as:

$$Z_1(T, L) = \frac{1}{h} \int_{-\infty}^{\infty} e^{-\beta \frac{p^2}{2m}} dp \int_0^L e^{\beta \alpha \ln\left(\frac{x}{x_0}\right)} dx \quad (8)$$

The first integral is

$$\int_{-\infty}^{\infty} e^{-\beta \frac{p^2}{2m}} dp = \sqrt{\frac{2m\pi}{\beta}} \quad (9)$$

The second integral is

$$\int_0^L e^{\beta \alpha \ln\left(\frac{x}{x_0}\right)} dx = \frac{L^{\alpha\beta+1}}{\alpha\beta+1} \frac{1}{x_0^{\alpha\beta}} \quad (10)$$

Then

$$Z_1(T, L) = \frac{1}{h} \sqrt{\frac{2m\pi}{\beta}} \frac{L^{\alpha\beta+1}}{\alpha\beta+1} \frac{1}{x_0^{\alpha\beta}} \quad (11)$$

- (b) Since the N particles are independent, we have $Z_N = \frac{1}{N!} Z_1^N$ and, therefore, $F = -k_B T N \ln Z_1 + K_b T (N \ln N - N)$ after applying Stirling's.

$$F(T, N, L) = -k_B T N \ln \left(\frac{1}{h} \sqrt{\frac{2m\pi}{\beta}} \frac{L^{\alpha\beta+1}}{\alpha\beta+1} \frac{1}{x_0^{\alpha\beta}} \right) + K_b T N (\ln N - 1) \quad (12)$$

- (c) The force Ψ that the particles exert against the ends of the segment is

$$\Psi = -\left.\frac{\partial F}{\partial L}\right|_{T,N} = \frac{k_B T N (\alpha\beta + 1)}{L} = \frac{k_B T N \left(\frac{\alpha}{k_B T} + 1\right)}{L} \quad (13)$$

- (d) We aim at finding the $\lim_{T \rightarrow 0} \Psi(T)$.

$$\lim_{T \rightarrow 0} \Psi = \lim_{T \rightarrow 0} \frac{k_B T N \left(\frac{\alpha}{k_B T} + 1\right)}{L} = \lim_{T \rightarrow 0} \frac{N\alpha + k_B T N}{L} = \frac{N\alpha}{L} \quad (14)$$

Problem 2:

Consider an ideal gas of particles with mass m in a 3D volume V in contact with a reservoir at temperature T and chemical potential μ . Each particle has an additional internal degree of freedom specified by a quantum number $m = -1, 0, 1$, with energy $\epsilon > 0$ for $m = 0$ and energy equal to zero otherwise.

- Compute the grand partition function of the gas.
- Determine the equation of state of the gas.
- Determine the average energy of the gas in terms of the expected number of particles $\langle N \rangle$. Identify the contribution from the internal degrees of freedom to the energy and discuss the behavior of this contribution for high and low temperature.

Solution Problem 2:

- Compute the grand partition function of the gas.

First we note that the Hamiltonian has an expression:

$$\mathcal{H} = \sum_i \frac{p_i^2}{2m} + \epsilon_{m,i} \quad (15)$$

where $\epsilon_{m,i} = 0, \epsilon$ depending on m . These are non-interacting particles so that the term $e^{-\beta\mathcal{H}}$ factorizes, which means that the canonical partition function factorizes as well $Z_{\text{canonical}} = \frac{1}{N!} \prod_i Z_i = \frac{Z_1^N}{N!}$, with Z_1 being the canonical partition function for one particle.

We obtain the partition function as

$$\mathcal{Z} = \sum_N z^N Z_{\text{canonical}}(N, T, V) = \sum_N z^N \frac{Z_1^N}{N!} = e^{zZ_1} \quad z = e^{\beta\mu} \quad (16)$$

where

$$Z_1 = \int \frac{d\vec{q}}{h^3} d\vec{p} e^{-\beta \frac{p^2}{2m}} \sum_{m=\{-1,0,1\}} e^{-\beta\epsilon_m} = \frac{V}{h^3} \left(\frac{2\pi m}{\beta} \right)^{3/2} (2 + e^{-\beta\epsilon}) \quad (17)$$

The final solution is then:

$$\mathcal{Z} = \exp \left(z \frac{V}{h^3} \left(\frac{2\pi m}{\beta} \right)^{3/2} (2 + e^{-\beta\epsilon}) \right) \quad (18)$$

- Determine the equation of state of the gas.

To determine the equation of state we first need to compute $\langle N \rangle$:

$$\langle N \rangle = \frac{1}{\beta} \frac{\partial \log Z}{\partial \mu} = Z_1 z = z \frac{V}{\hbar^3} \left(\frac{2\pi m}{\beta} \right)^{3/2} (2 + e^{-\beta\epsilon}) \quad (19)$$

Then we compute the grand free energy $\Phi = -k_B T \log \mathcal{Z}$ and use the equivalence $\Phi = -PV$ so that

$$PV = k_B T \log \mathcal{Z} = k_B T z Z_1 = k_B T \langle N \rangle, \quad (20)$$

which is the equation of state.

- (c) Determine the average energy of the gas in terms of the expected number of particles $\langle N \rangle$. Identify the contribution from the internal degrees of freedom to the energy and discuss the behavior of this contribution for high and low temperature.

To compute the average energy we use

$$\begin{aligned} \langle E \rangle &= - \left(\frac{\partial \log \mathcal{Z}}{\partial \beta} \right)_{\beta\mu} \\ &= \frac{3}{2\beta} z \frac{V}{\hbar^3} \left(\frac{2\pi m}{\beta} \right)^{3/2} (2 + e^{-\beta\epsilon}) + \frac{\epsilon e^{-\beta\epsilon}}{2 + e^{-\beta\epsilon}} z \frac{V}{\hbar^3} \left(\frac{2\pi m}{\beta} \right)^{3/2} (2 + e^{-\beta\epsilon}) \\ &= \frac{3}{2} \langle N \rangle k_B T + \frac{\epsilon e^{-\beta\epsilon}}{2 + e^{-\beta\epsilon}} \langle N \rangle \end{aligned} \quad (21)$$

The first term comes from the kinetic energy (equipartition theorem), the second term comes from the internal degree of freedom:

$$\langle E_{\text{int}} \rangle = \frac{\epsilon e^{-\beta\epsilon}}{2 + e^{-\beta\epsilon}} \langle N \rangle. \quad (22)$$

Limiting behavior At low temperature:

$$\lim_{T \rightarrow 0} \frac{\epsilon e^{-\beta\epsilon}}{2 + e^{-\beta\epsilon}} \langle N \rangle \rightarrow 0 \quad (23)$$

At high temperature:

$$\lim_{T \rightarrow \infty} \frac{\epsilon e^{-\beta\epsilon}}{2 + e^{-\beta\epsilon}} \langle N \rangle \rightarrow \frac{\epsilon}{3} \langle N \rangle \quad (24)$$